

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level
Advanced International Certificate of Education

MARK SCHEME for the June 2004 question papers

9709 MATHEMATICS

9709/01	Paper 1 (Pure 1), maximum raw mark 75
9709/02	Paper 2 (Pure 2), maximum raw mark 50
9709/03, 8719/03	Paper 3 (Pure 3), maximum raw mark 75
9709/04	Paper 4 (Mechanics 1), maximum raw mark 50
9709/05, 8719/05	Paper 5 (Mechanics 2), maximum raw mark 50
9709/06, 0390/06	Paper 6 (Probability and Statistics 1), maximum raw mark 50
9709/07, 8719/07	Paper 7 (Probability and Statistics 2), maximum raw mark 50

These mark schemes are published as an aid to teachers and students, to indicate the requirements of the examination. They show the basis on which Examiners were initially instructed to award marks. They do not indicate the details of the discussions that took place at an Examiners' meeting before marking began. Any substantial changes to the mark scheme that arose from these discussions will be recorded in the published *Report on the Examination*.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the *Report on the Examination*.

- CIE will not enter into discussion or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the June 2004 question papers for most IGCSE and GCE Advanced Level syllabuses.



UNIVERSITY of CAMBRIDGE
International Examinations

Grade thresholds taken for Syllabus 9709 (Mathematics) in the June 2004 examination.

	maximum mark available	minimum mark required for grade:		
		A	B	E
Component 1	75	63	56	31
Component 2	50	37	33	18
Component 3	75	61	55	29
Component 4	50	38	34	18
Component 5	50	36	32	17
Component 6	50	38	34	19
Component 7	50	42	37	22

The thresholds (minimum marks) for Grades C and D are normally set by dividing the mark range between the B and the E thresholds into three. For example, if the difference between the B and the E threshold is 24 marks, the C threshold is set 8 marks below the B threshold and the D threshold is set another 8 marks down. If dividing the interval by three results in a fraction of a mark, then the threshold is normally rounded down.

Mark Scheme Notes

- Marks are of the following three types:
 - M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
 - A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
 - B Mark for a correct result or statement independent of method marks.
- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

- The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only – often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

- **MR -1** A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
- **PA -1** This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

June 2004

GCE A AND AS LEVEL

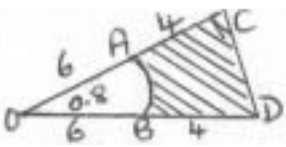
MARK SCHEME

MAXIMUM MARK: 75

SYLLABUS/COMPONENT: 9709/01

**MATHEMATICS
Paper 1 (Pure 1)**

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	1

1. (i) $a/(1-r) = 256$ and $a = 64$ $\rightarrow r = \frac{3}{4}$ (ii) $S_{10} = 64(1-0.75^{10}) / (1-0.75)$ $\rightarrow S_{10} = 242$	M1 A1 [2] M1 A1 [2]	Use of correct formula Correct only Use of correct formula – 0.75^{10} not 0.75^9 Correct only
2. $\int_0^1 \sqrt{3x+1} dx = (3x+1)^{1.5} \div 1.5$ then 3 $\rightarrow []$ at 1 – $[]$ at 0 $\rightarrow 16/9 - 2/9 = 14/9$ or 1.56	B1 M1 M1 A1 [4]	MI for $(3x+1)^{1.5} \div 1.5$ For division by 3 Must attempt $[]$ at $x=0$ (not assume it is 0) and be using an integrated function Fraction or decimal. (1.56+C loses this A1)
3. (i) $\sin^2 \theta + 3\sin \theta \cos \theta = 4\cos^2 \theta$ divides by $\cos^2 \theta$ $\rightarrow \tan^2 \theta + 3\tan \theta = 4$ (ii) Solution $\tan \theta = 1$ or $\tan \theta = -4$ $\rightarrow \theta = 45^\circ$ or 104.0°	M1 A1 [2] M1 A1 A1 [3]	Knowing to divide by $\cos^2 \theta$ Correct quadratic (not nec = 0) Correct solution of quadratic = 0 Correct only for each one.
4. (i) Coeff of $x^3 = 6C3 \times 2^3$ $= 160$ (ii) Term in $x^2 = 6C2 \times 2^2 = 60$ reqd coeff = $1 \times (i) - 3 \times 60$ $\rightarrow -20$	B1 B1 B1 [3] B1 M1 A1 [3]	B1 for $6C3$ B1 for 2^3 B1 for 160 B1 for 60 (could be given in (i)) Needs to consider 2 terms co
5.  (i) Area of sector = $\frac{1}{2} 6^2 0.8$ (14.4) Area of triangle = $\frac{1}{2} 10^2 \sin 0.8$ (35.9) $\rightarrow$ Shaded area = 21.5 (ii) Arc length = 6×0.8 (4.8) CD (by cos rule) or $2 \times 10 \sin 0.4$ (7.8) $\rightarrow$ Perimeter = $8 + 4.8 + 7.8 = 20.6$	M1 M1 A1 [3] M1 M1 A1 A1 [4]	Use of $\frac{1}{2}r^2\theta$ with radians Use of $\frac{1}{2}ab\sin C$ or $\frac{1}{2}bh$ with trig Correct only Use of $s=r\theta$ with radians Any correct method – allow if in (i) Correct only

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	1

6. (i) eliminates x (or y) completely $\rightarrow x^2+x-6=0$ or $y^2-17y+66=0$ Solution of quadratic = 0 $\rightarrow (2, 6)$ and $(-3, 11)$ (ii) Midpoint = $(-\frac{1}{2}, 8\frac{1}{2})$ Gradient of line = -1 Gradient of perpendicular = 1 $\rightarrow y - 8\frac{1}{2} = 1(x + \frac{1}{2})$ (or $y = x + 9$)	M1 A1 DM1 A1 [4] B1 ✓ M1 M1 A1 [4]	Needs x or y removed completely Correct only (no need for = 0) Equation must = 0. Everything ok. For his two points in (i) Use of y-step x-step (beware fortuitous) Use of $m_1m_2 = -1$ Any form – needs the M marks.
7. (i) Differentiate $y=18/x \rightarrow -18x^{-2}$ Gradient of tangent = $-\frac{1}{2}$ Gradient of normal = 2 Eqn of normal $y-3 = 2(x-6)$ $(y=2x-9)$ If $y = 0$, $x = 4\frac{1}{2}$ (ii) $\text{Vol} = \pi \int \frac{324}{x^2} dx = \pi[-324x^{-1}]$. Uses value at $x=6$ – value at $x= 4.5$ $-54\pi - -72\pi = 18\pi$	M1 A1 DM1 DM1 A1 [5] M1 A1 DM1 A1 [4]	Any attempt at differentiation For $-\frac{1}{2}$ Use of $m_1m_2 = -1$ Correct method for eqn of line Ans given – beware fortuitous answers. Use of $\int y^2 dx$ for M. correct(needs π) for A Use of 6 and 4.5 Beware fortuitous answers (ans given)
8. (i) $2h + 2r + \pi r = 8$ $\rightarrow h = 4 - r - \frac{1}{2}\pi r$  (ii) $A=2rh+\frac{1}{2}\pi r^2 \rightarrow A = r(8-2r-\pi r) + \frac{1}{2}\pi r^2$ $\rightarrow A = 8r - 2r^2 - \frac{1}{2}\pi r^2$ (iii) $dA/dr = 8 - 4r - \pi r$ $= 0$ when $r = 1.12$ (or $8/(4+\pi)$) (iv) $d^2A/dr^2 = -4 - \pi$ This is negative $\rightarrow$ Maximum	M1 A1 [2] M1 A1 M1 A1 DM1 A1 [4] M1 A1 [2]	Reasonable attempt at linking 4 lengths + correct formula for $\frac{1}{2}C$ or C. Co in any form with h subject. Adds rectangle + $\frac{1}{2}x$circle (eqn on own ok) Co beware fortuitous answers (ans given) Knowing to differentiate + some attempt Setting his dA/dr to 0. Decimal or exact ok. Looks at 2nd differential or other valid complete method. Correct deduction but needs d^2A/dr^2 correct.

Page 3	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	1

9. $\vec{OA} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}, \vec{OB} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}, \vec{OC} = \begin{pmatrix} 4 \\ 2 \\ p \end{pmatrix}, \vec{OD} = \begin{pmatrix} -1 \\ 0 \\ q \end{pmatrix}$ (i) $\vec{AB} = \mathbf{b} - \mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$ Unit vector = $(2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}) / \sqrt{(2^2 + 4^2 + 4^2)}$ $= \pm (2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}) / 6$ (ii) $\vec{OA} \cdot \vec{OC} = 4 + 6 - p$ $= 0$ for 90° $\rightarrow p = 10$ (iii) $(-2)^2 + 3^2 + (q+1)^2 = 7^2$ $\rightarrow (q+1)^2 = 36$ or $q^2 + 2q = 35$ $q = 5$ and $q = -7$	M1 M1 A1 [3] M1 DM1 A1 [3] M1 A1 DM1 A1 or B1 B1 [4]	Condone notation throughout. Allow column vectors or $\mathbf{i}, \mathbf{j}, \mathbf{k}$ throughout Use of $\mathbf{b} - \mathbf{a}$, rather than $\mathbf{b} + \mathbf{a}$ or $\mathbf{a} - \mathbf{b}$ Dividing by the modulus of "his" $\vec{AB}$ Co (allow – for candidates using $\mathbf{a} - \mathbf{b}$) Use of $x_1x_2 + y_1y_2 + z_1z_2$ Setting to 0 + attempt to solve co Correct method for length with $\pm \mathbf{d} - \mathbf{a}, \mathbf{d} + \mathbf{a}$ Correct quadratic equation Correct method of solution. Both correct. Or B1 for each if $(q+1)^2 = 36$, $q = 5$ only.
10. $f: x \mapsto x^2 - 2x, \quad g: x \mapsto 2x + 3$ (i) $x^2 - 2x - 15 = 0$ End-points -3 and 5 $\rightarrow x < -3$ and $x > 5$ (ii) Uses $dy/dx = 2x - 2 = 0$ or $(x-1)^2 - 1$ Minimum at $x = 1$ or correct form Range of y is $f(x) \geq -1$ No inverse since not 1 : 1 (or equivalent) (iii) $gf(x) = 2(x^2 - 2x) + 3 \quad (2x^2 - 4x + 3)$ $b^2 - 4ac = 16 - 24 = -8 \rightarrow -ve$ $\rightarrow$ No real solutions. [or $gf(x) = 0 \rightarrow f(x) = -3/2$. Imposs from (ii)] (iv) $y = 2x + 3$ correct line on diagram Either inverse as mirror image in $y = x$ or $y = g^{-1}(x) = \frac{1}{2}(x - 3)$ drawn	M1 A1 A1 [3] M1 A1 A1 B1 [4] M1 M1 A1 [3] B2, 1, 0 [2]	Equation set to 0 and solved. Correct end-points, however used Co-inequalities – not $\leq$ or $\geq$ Any valid complete method for x value Correct only Correct for his value of "x" – must be $\geq$ Any valid statement. Must be gf not fg – for unsimplified ans. Used on quadratic = 0, even if fg used. Must be using gf and correct assumption and statement needed. 3 things needed – B1 if one missing. • g correct, • g^{-1} correct – not parallel to g • $y = x$ drawn or statement re symmetry
DM1 for quadratic equation. Equation must be set to 0. Formula $\rightarrow$ must be correct and correctly used – allow for numerical errors though in b^2 and $-4ac$. Factors $\rightarrow$ attempt to find 2 brackets. Each bracket then solved to 0.		

June 2004

GCE AS LEVEL

MARK SCHEME

MAXIMUM MARK: 50

SYLLABUS/COMPONENT: 9709/02

**MATHEMATICS
Paper 2 (Pure 2)**

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	2

- 1 Use logarithms to linearise an equation M1
Obtain $\frac{x}{y} = \frac{\ln 5}{\ln 2}$ or equivalent A1
Obtain answer 2.32 A1 **3**
- 2 (i) Use the given iterative formula correctly at least ONCE with $x_1 = 3$ M1
Obtain final answer 3.142 A1
Show sufficient iterations to justify its accuracy to 3 d.p. A1 **3**
- (ii) State any suitable equation e.g. $x = \frac{1}{5} \left(4x + \frac{306}{x^4} \right)$ B1
Derive the given answer α (or x) = $\sqrt[5]{306}$ B1 **2**
- 3 (i) Substitute $x = 3$ and equate to zero M1
Obtain answer $\alpha = -1$ A1 **2**
- (ii) At any stage, state that $x = 3$ is a solution B1
EITHER: Attempt division by $(x-3)$ reaching a partial quotient of $2x^2 + kx$ M1
Obtain quadratic factor $2x^2 + 5x + 2$ A1
Obtain solutions $x = -2$ and $x = -\frac{1}{2}$ A1
OR: Obtain solution $x = -2$ by trial and error B1
Obtain solution $x = -\frac{1}{2}$ similarly B2 **4**
[If an attempt at the quadratic factor is made by inspection, the M1 is earned if it reaches an unknown factor of $2x^2 + bx + c$ and an equation in b and/or c .]
- 4 (i) State answer $R = 5$ B1
Use trigonometric formulae to find α M1
Obtain answer $\alpha = 53.13^\circ$ A1 **3**
- (ii) Carry out, or indicate need for, calculation of $\sin^{-1}(4.5/5)$ M1
Obtain answer 11.0° A1√
Carry out correct method for the second root e.g. $180^\circ - 64.16^\circ - 53.13^\circ$ M1
Obtain answer 62.7° and no others in the range A1√ **4**
[Ignore answers outside the given range.]
- (iii) State least value is 2 B1√ **1**
- 5 (i) State derivative of the form $(e^x \pm xe^x)$. Allow $xe^x \pm e^x$ {via quotient rule} M1
Obtain correct derivative of $e^{\pm x} - xe^x$ A1
Equate derivative to zero and solve for x M1
Obtain answer $x = 1$ A1 **4**
- (ii) Show or imply correct ordinates 0, 0.367879..., 0.27067... B1
Use correct formula, or equivalent, with $h = 1$ and three ordinates M1
Obtain answer 0.50 with no errors seen A1 **3**
- (iii) Justify statement that the rule gives an under-estimate B1 **1**

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	2

- 6 (i) State that $\frac{dx}{dt} = 2 + \frac{1}{t}$ or $\frac{dy}{dt} = 1 - \frac{4}{t^2}$, or equivalent B1
- Use $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ M1
- Obtain the given answer A1 **3**
- (ii) Substitute $t = 1$ in $\frac{dy}{dx}$ and both parametric equations M1
- Obtain $\frac{dy}{dx} = -1$ and coordinates (2, 5) A1
- State equation of tangent in any correct horizontal form e.g. $x + y = 7$ A1√ **3**
- (iii) Equate $\frac{dy}{dx}$ to zero and solve for t M1
- Obtain answer $t = 2$ A1
- Obtain answer $y = 4$ A1
- Show by any method (but not via $\frac{d}{dt}(y')$) that this is a minimum point A1 **4**
- 7 (i) Make relevant use of the $\cos(A + B)$ formula M1*
- Make relevant use of $\cos 2A$ and $\sin 2A$ formulae M1*
- Obtain a correct expression in terms of $\cos A$ and $\sin A$ A1
- Use $\sin^2 A = 1 - \cos^2 A$ to obtain an expression in terms of $\cos A$ M1(dep*)
- Obtain given answer correctly A1 **5**
- (ii) Replace integrand by $\frac{1}{4} \cos 3x + \frac{3}{4} \cos x$, or equivalent B1
- Integrate, obtaining $\frac{1}{12} \sin 3x + \frac{3}{4} \sin x$, or equivalent B1 + B1√
- Use limits correctly M1
- Obtain given answer A1 **5**

June 2004

GCE A AND AS LEVEL

MARK SCHEME

MAXIMUM MARK: 75

SYLLABUS/COMPONENT: 9709/03, 8719/03

**MATHEMATICS AND HIGHER MATHEMATICS
Paper 3 (Pure 3)**

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

- 1 Show correct sketch for $0 \leq x < \frac{1}{2}\pi$ B1
 Show correct sketch for $\frac{1}{2}\pi < x < \frac{3}{2}\pi$ or $\frac{3}{2}\pi < x \leq 2\pi$ B1
 Show completely correct sketch B1 **3**
 [SR: for a graph with $y = 0$ when $x = 0, \pi, 2\pi$ but otherwise of correct shape, award B1.]
- 2 *EITHER:* State or imply non-modular inequality $(2x+1)^2 < x^2$ or corresponding quadratic equation or pair of linear equations $(2x+1) = \pm x$ B1
 Expand and make a reasonable solution attempt at a 3-term quadratic, or solve two linear equations M1
 Obtain critical values $x = -1$ and $x = -\frac{1}{3}$ only A1
 State answer $-1 < x < -\frac{1}{3}$ A1
OR: Obtain the critical value $x = -1$ from a graphical method, or by inspection, or by solving a linear inequality or equation B1
 Obtain the critical value $x = -\frac{1}{3}$ (deduct B1 from B3 if extra values are obtained) B2
 State answer $-1 < x < -\frac{1}{3}$ B1 **4**
 [Condone $\leq$ for $<$; accept -0.33 for $-\frac{1}{3}$.]
- 3 *EITHER:* State $6y \frac{dy}{dx}$ as the derivative of $3y^2$ B1
 State $\pm 4x \frac{dy}{dx} \pm 4y$ as the derivative of $-4xy$ B1
 Equate attempted derivative of LHS to zero and solve for $\frac{dy}{dx}$ M1
 Obtain answer 2 A1
 [The M1 is conditional on at least one of the B marks being obtained. Allow any combination of signs for the second B1.]
OR: Obtain a correct expression for y in terms of x B1
 Differentiate using chain rule M1
 Obtain derivative in any correct form A1
 Substitute $x = 2$ and obtain answer 2 only A1 **4**
 [The M1 is conditional on a reasonable attempt at solving the quadratic in y being made.]

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

- 4 (i) State or imply $2^{-x} = \frac{1}{y}$ B1
Obtain 3-term quadratic e.g. $y^2 - y - 1 = 0$ B1 2
- (ii) Solve a 3-term quadratic, obtaining 1 or 2 roots M1
Obtain answer $y = (1 + \sqrt{5})/2$, or equivalent A1
Carry out correct method for solving an equation of the form $2^x = a$, where $a > 0$, reaching a ratio of logarithms M1
Obtain answer $x = 0.694$ only A1 4
- 5 (i) Make relevant use of formula for $\sin 2\theta$ or $\cos 2\theta$ M1
Make relevant use of formula for $\cos 4\theta$ M1
Complete proof of the given result A1 3
- (ii) Integrate and obtain $\frac{1}{8}(\theta - \frac{1}{4}\sin 4\theta)$ or equivalent B1
Use limits correctly with an integral of the form $a\theta + b\sin 4\theta$, where $ab \neq 0$ M1
Obtain answer $\frac{1}{8}(\frac{1}{3}\pi + \frac{\sqrt{3}}{8})$, or exact equivalent A1 3
- 6 Separate variables and attempt to integrate M1
Obtain terms $\frac{1}{3}\ln(y^3 + 1)$ and x , or equivalent A1 + A1
Evaluate a constant or use limits $x = 0$, $y = 1$ with a solution containing terms $k\ln(y^3 + 1)$ and x , or equivalent M1
Obtain any correct form of solution e.g. $\frac{1}{3}\ln(y^3 + 1) = x + \frac{1}{3}\ln 2$ A1✓
Rearrange and obtain $y = (2e^{3x} - 1)^{\frac{1}{3}}$, or equivalent A1 6
[f.t. is on $k \neq 0$.]
- 7 (i) Evaluate cubic when $x = -1$ and $x = 0$ M1
Justify given statement correctly A1 2
[If calculations are not given but justification uses correct statements about signs, award B1.]
- (ii) State $x = \frac{2x^3 - 1}{3x^2 + 1}$, or equivalent B1
Rearrange this in the form $x^3 + x + 1 = 0$ (or vice versa) B1 2

Page 3	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

- (iii) Use the iterative formula correctly at least once M1
 Obtain final answer -0.68 A1
 Show sufficient iterations to justify its accuracy to 2d.p., or show there is a sign change in the interval $(-0.685, -0.675)$ A1 **3**

- 8 (i) *EITHER*: Solve the quadratic and use $\sqrt{-1} = i$ M1
 Obtain roots $\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $\frac{1}{2} - i\frac{\sqrt{3}}{2}$ or equivalent A1
OR: Substitute $x + iy$ and solve for x or y M1
 Obtain correct roots A1 **2**

- (ii) State that the modulus of each root is equal to 1 B1✓
 State that the arguments are $\frac{1}{3}\pi$ and $-\frac{1}{3}\pi$ respectively B1✓ + B1✓ **3**
 [Accept degrees and $\frac{5}{3}\pi$ instead of $-\frac{1}{3}\pi$. Accept a modulus in the form $\sqrt{\frac{p}{q}}$ or $\sqrt{n}$, where p, q, n are integers. An answer which only gives roots in modulus-argument form earns B1 for both the implied moduli and B1 for both the implied arguments.]

- (iii) *EITHER*: Verify $z^3 = -1$ for each root B1 + B1
OR: State $z^3 + 1 = (z + 1)(z^2 - z + 1)$ B1
 Justify the given statement B1
OR: Obtain $z^3 = z^2 - z$ B1
 Justify the given statement B1 **2**

- 9 (i) State or imply $f(x) \equiv \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+1}$ B1
EITHER: Use any relevant method to obtain a constant M1
 Obtain one of the values: $A = -1, B = 4$ and $C = -2$ A1
 Obtain the remaining two values A1
OR: Obtain one value by inspection B1
 State a second value B1
 State the third value B1 **4**
 [Apply the same scheme to the form $\frac{A}{x-2} + \frac{Bx+C}{x^2-1}$ which has $A = 4, B = -3$ and $C = 1$.]

Page 4	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

- (ii) Use correct method to obtain the first two terms of the expansion of $(x-1)^{-1}$ or $(x-2)^{-1}$ or $(x+1)^{-1}$ M1

Obtain any correct unsimplified expansion of the partial fractions up to the terms in x^3

(deduct A1 for each incorrect expansion)

A1✓ + A1✓ + A1✓

Obtain the given answer correctly

A1 5

[Binomial coefficients involving -1 , e.g. $\binom{-1}{1}$, are not sufficient for the M1 mark. The f.t. is on A, B, C.]

[Apply a similar scheme to the alternative form of fractions in (i), awarding M1*A1✓A1✓ for the expansions, M1(dep*) for multiplying by $Bx + C$, and A1 for obtaining the given answer correctly.]

[In the case of an attempt to expand $(x^2 + 7x - 6)(x-1)^{-1}(x-2)^{-1}(x+1)^{-1}$, give M1A1A1A1 for the expansions and A1 for multiplying out and obtaining the given answer correctly.]

[Allow attempts to multiply out $(x-1)(x-2)(x+1)(-3+2x-\frac{3}{2}x^2+\frac{11}{4}x^3)$, giving B1 for reduction to a product of two expressions correct up to their terms in x^3 , M1 for attempting to multiply out at least as far as terms in x^2 , A1 for a correct expansion up to terms in x^3 , and A1 for correctly obtaining the answer $x^2 + 7x - 6$ and also showing there is no term in x^3 .]

[Allow the use of Maclaurin, giving M1A1✓ for $f(0) = -3$ and $f'(0) = 2$, A1✓ for $f''(0) = -3$, A1✓ for $f'''(0) = \frac{33}{2}$, and A1 for obtaining the given answer correctly (f.t. is on A, B, C if used).]

- 10 (i) State x-coordinate of A is 1

B1 1

- (ii) Use product or quotient rule

M1

Obtain derivative in any correct form e.g. $-\frac{2\ln x}{x^3} + \frac{1}{x} \cdot \frac{1}{x^2}$

A1

Equate derivative to zero and solve for $\ln x$

M1

Obtain $x = e^{\frac{1}{2}}$ or equivalent (accept 1.65)

A1

Obtain $y = \frac{1}{2e}$ or exact equivalent not involving $\ln$

A1 5

[SR: if the quotient rule is misused, with a 'reversed' numerator or x^2 instead of x^4 in the denominator, award M0A0 but allow the following M1A1A1.]

- (iii) Attempt integration by parts, going the correct way

M1

Obtain $-\frac{\ln x}{x} + \int \frac{1}{x} \cdot \frac{1}{x} dx$ or equivalent

A1

Obtain indefinite integral $-\frac{\ln x}{x} - \frac{1}{x}$

A1

Use x-coordinate of A and e as limits, having integrated twice

M1

Obtain exact answer $1 - \frac{2}{e}$, or equivalent

A1 5

[If $u = \ln x$ is used, apply an analogous scheme to the result of the substitution.]

Page 5	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

11 (i) EITHER: Obtain a vector in the plane e.g. $\overrightarrow{PQ} = -3\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ B1

Use scalar product to obtain a relevant equation in a, b, c e.g. $-3a + 4b + c = 0$ or

$6a - 2b + c = 0$ or $3a + 2b + 2c = 0$ M1

State two correct equations in a, b, c A1

Solve simultaneous equations to obtain one ratio e.g. $a : b$ M1

Obtain $a : b : c = 2 : 3 : -6$ or equivalent A1

Obtain equation $2x + 3y - 6z = 8$ or equivalent A1

[The second M1 is also given if say c is given an arbitrary value and a or b is found.

The following A1 is then given for finding the correct values of a and b .]

OR: Substitute for P, Q, R in equation of plane and state 3 equations in a, b, c, d B1

Eliminate one unknown, e.g. d , entirely M1

Obtain 2 equations in 3 unknowns A1

Solve to obtain one ratio e.g. $a : b$ M1

Obtain $a : b : c = 2 : 3 : -6$ or equivalent A1

Obtain equation $2x + 3y - 6z = 8$ or equivalent A1

[The first M1 is also given if say d is given an arbitrary value and two equations in

two unknowns, e.g. a and b , are obtained. The following A1 is for two correct

equations. Solving to obtain one unknown earns the second M1 and the following

A1 is for finding the correct values of a and b .]

OR: Obtain a vector in the plane e.g. $\overrightarrow{QR} = 6\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ B1

Find a second vector in the plane and form correctly a 2-parameter equation for the plane M1

Obtain equation in any correct form e.g. $\mathbf{r} = \lambda(-3\mathbf{i} + 4\mathbf{j} + \mathbf{k}) + \mu(6\mathbf{i} - 2\mathbf{j} + \mathbf{k}) + \mathbf{i} - \mathbf{k}$ A1

State 3 equations in x, y, z, λ , and μ A1

Eliminate λ and μ M1

Obtain equation $2x + 3y - 6z = 8$ or equivalent A1

OR: Obtain a vector in the plane e.g. $\overrightarrow{PR} = 3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ B1

Obtain a second vector in the plane and calculate the vector product of the two vectors, e.g. $(-3\mathbf{i} + 4\mathbf{j} + \mathbf{k}) \times (3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$ M1

Obtain 2 correct components of the product A1

Obtain correct product e.g. $6\mathbf{i} + 9\mathbf{j} - 18\mathbf{k}$ or equivalent A1

Substitute in $2x + 3y - 6z = d$ and find d or equivalent M1

Obtain equation $2x + 3y - 6z = 8$ or equivalent A1

6

Page 6	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	3

- (ii) *EITHER*: State equation of SN is $\mathbf{r} = 3\mathbf{i} + 5\mathbf{j} - 6\mathbf{k} + \lambda(2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k})$ or equivalent B1✓
Express x, y, z in terms of λ e.g. $(3 + 2\lambda, 5 + 3\lambda, -6 - 6\lambda)$ B1✓
Substitute in the equation of the plane and solve for λ M1
Obtain $\overrightarrow{ON} = \mathbf{i} + 2\mathbf{j}$, or equivalent A1
Carry out method for finding SN M1
Show that $SN = 7$ correctly A1

- OR*: Letting $\overrightarrow{ON} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, obtain two equations in x, y, z by equating scalar
product of $\overrightarrow{NS}$ with two of $\overrightarrow{PQ}, \overrightarrow{QR}, \overrightarrow{RP}$ to zero B1✓+ B1✓
Using the plane equation as third equation, solve for x, y , and z M1
Obtain $\overrightarrow{ON} = \mathbf{i} + 2\mathbf{j}$, or equivalent A1
Carry out method for finding SN M1
Show that $SN = 7$ correctly A1

- OR*: Use Cartesian formula or scalar product of $\overrightarrow{PS}$ with a normal vector to find SN M1
Obtain $SN = 7$ A1
State a unit normal $\hat{\mathbf{n}}$ to the plane B1✓
Use $\overrightarrow{ON} = \overrightarrow{OS} \pm 7\hat{\mathbf{n}}$ M1
Obtain an unsimplified expression e.g. $3\mathbf{i} + 5\mathbf{j} - 6\mathbf{k} \pm 7(\frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} - \frac{6}{7}\mathbf{k})$ A1✓
Obtain $\overrightarrow{ON} = \mathbf{i} + 2\mathbf{j}$, or equivalent, only A1

6

June 2004

GCE A AND AS LEVEL

MARK SCHEME

MAXIMUM MARK: 50

SYLLABUS/COMPONENT: 9709/04

**MATHEMATICS
Paper 4 (Mechanics 1)**

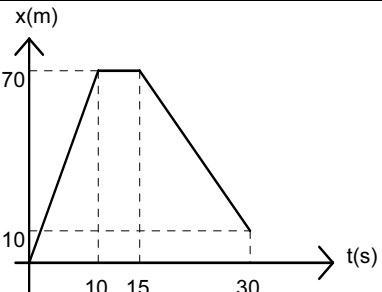
Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	4

1	(i)	$F = 13 \cos \alpha$ Frictional component is 12 N	M1 A1 2	For resolving forces horizontally
	(ii)	$R = 1.1 \times 10 + 13 \sin \alpha$ Normal component is 16 N	M1 A1 2	For resolving forces vertically (3 terms needed)
	(iii)	Coefficient of friction is 0.75	B1 ft 1	

2		$X = 100 + 250 \cos 70^\circ$ $Y = 300 - 250 \sin 70^\circ$ $R^2 = 185.5^2 + 65.1^2$ $R = 197$ $\tan \alpha = 65.1/185.5$ $\alpha = 19.3$	B1 B1 M1 A1 ft M1 A1 ft 6	For using $R^2 = X^2 + Y^2$ ft only if one B1 is scored or if the expressions for the candidate's X and Y are those of the equilibrant For using $\tan \alpha = Y/X$ ft only if one B1 is scored SR for sin/cos mix (max 4/6) $X = 100 + 250 \sin 70^\circ$ and $Y = 300 - 250 \cos 70^\circ$ (334.9 and 214.5) B1 Method marks as scheme M1 M1 $R = 398 \text{ N}$ and $\alpha = 32.6$ A1
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OR

		316(.227766..) or 107(.4528..) or 299(.3343..) or 71.565 ...° or 37.2743 ...° or -51.7039 ...° $R^2 = 316.2^2 + 250^2 - 2 \times 316.2 \times 250 \cos 38.4^\circ$ $R^2 = 107.5^2 + 100^2 - 2 \times 107.5 \times 100 \cos 142.7^\circ$ $R^2 = 299.3^2 + 300^2 - 2 \times 299.3 \times 300 \cos 38.3^\circ$ $R = 197$ $\sin(71.6 - \alpha) = 250 \sin 38.4 \div 197$ $\sin(37.3 - \alpha) = 100 \sin 142.7 \div 197$ $\sin(51.7 + \alpha) = 300 \sin 38.3 \div 197$ $\alpha = 19.3^\circ$	B1 B1 M1 A1 ft M1 A1 ft	Magnitude of the resultant of two of the forces Direction of the resultant of two of the forces For using the cosine rule to find R ft only if one B1 is scored For using the sine rule to find α ft only if one B1 is scored
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3	(i)	Distance AC is 70 m $7 \times 10 - 4 \times 15$ Distance AB is 10 m	B1 M1 A1 3	For using $ AB = AC - BC $
	(ii)		M1 A1 A1 ft 3	Graph consists of 3 connected straight line segments with, in order, positive, zero and negative slopes. $x(t)$ is single valued and the graph contains the origin 1 st line segment appears steeper than the 3 rd and the 3 rd line segment does not terminate on the t -axis Values of t (10, 15 and 30) and x (70, 70, 10) shown, or can be read without ambiguity from the scales SR (max 1 out of 3 marks) For first 2 segments correct B1

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	4

4	(i)	$KE = 0.2g(0.7)$ Kinetic energy is 1.4 J	M1 A1 2	For using $KE = PE$ lost and PE lost = mgh
	(ii)	$R = 0.2 \times 10 \times \cos 16.3^\circ$ $F = 0.288 \text{ N}$ $WD = 0.72 \text{ J}$ or $a = 1.36$ or resultant downward force = 0.272 N $KE = 1.4 - 0.72$ or $KE = \frac{1}{2} 0.2(2 \times 1.36 \times 2.5)$ or 0.272×2.5 Kinetic energy is 0.68 J	B1 B1 ft B1 ft M1 A1 ft 5	1.92 From $0.15R$ (may be implied by subsequent exact value 0.72, 1.36 or 0.68) From $2.5F$ or from $0.2a = 0.2 \times 10 \times (7/25) - F$ (may be implied by subsequent exact value 0.68) For using $KE = PE$ lost – WD or $KE = \frac{1}{2} mv^2$ and $v^2 = 2as$ or $KE = \text{resultant downward force} \times 2.5$

5	(i)	$10t^2 - 0.25t^4$ (+C) Expression is $10t^2 - 0.25t^4 - 36$	M1 DM1 A1 3	For integrating v For including constant of integration and attempting to evaluate it
	(ii)	Displacement is 60 m	A1 ft 1	Dependent on both M marks in (i); ft if there is not more than one error in $s(t)$
	(iii)	$(t^2 - 36)(1 - 0.25t^2) = 0$ Roots of quadratic are 4, 36 $t = 2, 6$	M1 A1 A1 ft 3	For attempting to solve $s = 0$ (depends on both method marks in (i)) or $\int_0^t v dt = 36$ (but not -36) for t^2 by factors or formula method ft only from 3 term quadratic in t^2

6	(i)	$DF - 400 = 1200 \times 0.5$ $20000 = 1000v$ Speed is 20 ms^{-1}	M1 A1 M1 A1 4	For using Newton's 2 nd law (3 terms needed) For using $P = Fv$
	(ii)	$20000/v - 400 = 0$ $v_{\max} = 50 \text{ ms}^{-1}$	M1 A1 2	For using $P = Fv$ and Newton's 2 nd law with $a = 0$ and $F = 400$ AG
	(iii)	$20000 = \frac{1500000}{\Delta T}$ or distance = $1500\,000/400 = 3750$ and time = $3750/50$ Time taken is 75 s	M1 A1 2	For using $P = \frac{\Delta W}{\Delta T}$ or for using 'distance = work done/400' and 'time = distance/50'

Page 3	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709	4

7	(i)	$25 = 30t - 5t^2 \Rightarrow t^2 - 6t + 5 = 0 \Rightarrow (t-1)(t-5) = 0$ or $v^2 = 30^2 - 500; t_{\text{up}} = (20 - 0)/10$ $t = 1, 5$ or $t_{\text{up}} = 2$ Time = $5 - 1 = 4$ s or Time = $2 \times 2 = 4$ s or $1 < t < 5$	M1 A1 A1 3	For using $25 = ut - \frac{1}{2}gt^2$ and attempting to solve for t or for using $v^2 = u^2 - 2g(25)$ and $t_{\text{up}} = (v - 0)/g$
	(ii)	$s_1 = 30t - 5t^2$ and $s_2 = 10t - 5t^2$ $30t - 10t = 25$ $t = 1.25$ $v_1 = 30 - 10 \times 1.25$ or $v_2 = 10 - 10 \times 1.25$ or $v_1^2 = 30^2 - 2 \times 10(29.6875)$ or $v_2^2 = 10^2 - 2 \times 10(4.6875)$ Velocities 17.5ms^{-1} and -2.5ms^{-1}	M1 M1 A1 M1 A1 5	For using $s = ut - \frac{1}{2}gt^2$ for P_1 and P_2 For using $s_1 = s_2 + 25$ and attempting to solve for t For using $v = u - gt$ (either case) or for calculating s_1 and substituting into $v_1^2 = 30^2 - 2 \times 10s_1$ or calculating s_2 and substituting into $v_2^2 = 10^2 - 2 \times 10s_2$

OR

	(ii)	$v_1 = 30 - 10t, v_2 = 10 - 10t$ $\Rightarrow v_1 - v_2 = 20$ $(30^2 - v_1^2) \div 20 = (10^2 - v_2^2) \div 20 + 25$ $v_1 - v_2 = 20, v_1^2 - v_2^2 = 300$ Velocities are 17.5 ms^{-1} and -2.5 ms^{-1}	M1 M1 A1 M1 A1 5	For using $v = u - gt$ for P_1 and P_2 and eliminating t For using $v^2 = u^2 - 2gs$ for P_1 and P_2 and then $s_1 = s_2 + 25$ For solving simultaneous equations in v_1 and v_2
	(iii)	$t_{\text{up}} = 3$ $3 - 1.25$ Time is 1.75 s or $1.25 < t < 3$	B1 M1 A1 3	For using t_{up} and above = $t_{\text{up}} - t_{\text{equal}}$

OR

	(iii)	$0 = 17.5 - 10t$ Time is 1.75 s or $1.25 < t < 3$	M2 A1	For using $0 = u - gt$ with u equal to the answer found for v_1 in (ii) SR (max 1 out of 3 marks) $0 = 17.5 + 10t$ B1 ft
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June 2004

GCE A AND AS LEVEL

MARK SCHEME

MAXIMUM MARK: 50

SYLLABUS/COMPONENT: 9709/05, 8719/05

**MATHEMATICS AND HIGHER MATHEMATICS
Paper 5 (Mechanics 2)**

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	5

Mechanics 2

- 1** For taking moments about the edge of the platform M1
- $(75g \times 0.9 = 25g \times x + 10g \times 1.1)$ (3 term equation)
- Two terms correct (unsimplified) A1
- Completely correct (unsimplified) A1
- Distance $MC = 3.16\text{m}$ A1 **4**
- NB:** If moments taken about other points, the force of the platform on the plank must be present at the edge of the platform for M1
- 2 (i)** Evaluates $\frac{2r \sin \alpha}{3\alpha} \times \cos \frac{\pi}{4}$ M1
- Obtains given answer correctly A1 **2**
- (ii)** For taking moments about AB M1
- $\{(5 \times 10 + \frac{1}{4}\pi 5^2) \bar{x} = (5 \times 10) \times 5 + \frac{1}{4}\pi 5^2(10 + \frac{20}{3\pi})\}$
- For the total area correct and the moment of the rectangle correct (unsimplified) A1
- For the moment of CDE correct (unsimplified) A1
- Distance is 7.01 cm A1 **4**
- 3** For applying Newton's 2nd law and using $a = v \frac{dv}{dx}$ M1
- $0.6v \frac{dv}{dx} = -\frac{3}{x^3}$ A1
- For separating the variables and integrating M1
- $0.3v^2 = -\frac{3x^{-2}}{(-2)}$ (+C) A1 ft
- (ft omission of minus sign in line 2 only)
- For using $v = 0$ when $x = 10$ M1
- $v^2 = \frac{5}{x^2} - \frac{1}{20}$ (aef) A1 ft
- (ft wrong sign in line 4 only)
- Speed is $\frac{\sqrt{3}}{2} \text{ms}^{-1}$ (=0.866) A1 **7**

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	5

- 4 (i) Distance of the rod from the hinge is $\frac{2.4}{2.5}(0.7)$ or $0.7\cos 16.26^\circ (=0.672)$ B1
[May be implied in moment equation]
For taking moments about the hinge (3 term equation) M1
 $0.672F = 68 \times 1.2 + 750 \times 2.4$ A1 ft
Force is 2800 N A1 **4**
- (ii) $X = 784$ (ft for $0.28F$) B1 ft
For resolving vertically (4 term equation) M1
 $Y = 1870$ (ft for $0.96F - 818$) A1 ft **3**

SR: For use of 680 N for weight of the beam: (i) B1, M1, A0. In (ii) ft 680, so 3/3 possible.

- 5 (i) For using $EPE = \frac{\lambda x^2}{2L}$ M1
 $EPE \text{ gain} = 2\left(\frac{200x^2}{2 \times 4}\right)$ ($=50x^2$) A1
 $GPE \text{ loss} = 10g(4 + x)$ B1
For using the principle of conservation of energy to form an equation M1
containing EPE, GPE and KE terms
 $[\frac{1}{2}10^2 + 50x^2 = 10g(4 + x)]$
Given answer obtained correctly A1 **5**

ALTERNATIVE METHOD:

- $T = \frac{200x}{4}$ B1
 $100 - 2\left(\frac{200x}{4}\right) = 10v \frac{dv}{dx}$ M1
 $\frac{1}{2}v^2 = 10x - 5x^2$ (+C) A1
Use $x = 0, v^2 = 8g$ M1
 $v^2 = 10(8 + 2x - x^2)$ A1
- (ii) For using $v = 0$ and factorizing or using formula method for solving M1
 $x = 4$ (only) A1 **2**

Page 3	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	5

- 6 (i) $2 = VT \sin 35^\circ - 5T^2$ or $2 = 25 \tan 35^\circ - \frac{25^2 \times 10}{2V^2 \cos^2 35^\circ}$ B1
- $25 = VT \cos 35^\circ$ B1
- For obtaining V^2 or T^2 in $AV^2 = B$ or $CT^2 = D$ form where A, B, C, D are numerical M1
- $[(25 \tan 35^\circ - 2) \cos^2 35^\circ] V^2 = 3125$ (aef) or
- $5T^2 = 25 \tan 35^\circ - 2$ (aef)]
- $V = 17.3$ or $T = 1.76$ A1
- $T = 1.76$ or $V = 17.3$ (ft $VT = 30.519365$) B1 ft 5
- (ii) For using $\dot{y} = V \sin 35^\circ - gT$ (must be component of V for M1) M1
- $\dot{y}_M (= 9.94 - 17.61 = -7.67) < 0 \rightarrow$ moving downwards A1 ft
- (ft on V and T)
- For using $v_M^2 = (V \cos 35^\circ)^2 + \dot{y}_M^2$ M1
- $(v_M^2 = ((14.20)^2 + (-7.67)^2)$ or
- For using the principle of conservation of energy
- $(\frac{1}{2} m (v_M^2 - 17.3^2) = -mg \times 2)$
- $v_M = 16.1 \text{ ms}^{-1}$ A1 4

LINES 1 AND 2 ALTERNATIVE METHODS

EITHER Compare 25 with $\frac{1}{2} R \left(\frac{v^2 \sin 70^\circ}{g} \right)$ M1

$25 > 14.1 \rightarrow$ moving downwards A1

OR Compare 1.76 with time to greatest height $\left(\frac{V \sin 35^\circ}{g} \right)$ M1

$1.76 > 0.994 \rightarrow$ moving downwards A1

OR $\frac{dy}{dx} = \tan 35^\circ - \frac{g \cdot 10}{V^2 \cos^2 35^\circ} (= -0.54)$ used M1

As $\tan \phi$ is negative $\rightarrow$ moving downwards A1

Page 4	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	5

- 7 (i) $T \cos 60^\circ = 0.5g$ ($T = 10$) B1
- For applying Newton's 2nd law horizontally and using $a = \frac{v^2}{r}$ M1
- (must be a component of T for M1)
- $T \sin 60^\circ = \frac{0.5v^2}{0.15 \sin 60^\circ}$ (for an equation in V^2) A1
- For substituting for T M1
- $= 1.5$ A1 **5**

ALTERNATIVELY:

- $a = \frac{v^2}{0.15 \sin 60^\circ}$ B1
- For applying Newton's 2nd law perpendicular to the string M1
- $0.5g \cos 30^\circ = 0.5(a \cos 60^\circ)$ A1
- For substituting for a M1
- $(5 \cos 30^\circ = 0.5^2 / 0.15 \tan 60^\circ)$ (for an equation in V^2)
- $= 1.5$ A1
- (ii) (a) $T \sin 45^\circ = \frac{0.5(0.9)^2}{0.15 \sin 45^\circ}$ B1
- Tension is 5.4 N B1 **2**
- (b) For resolving forces vertically M1
- $5.4 \cos 45^\circ + R = 0.5g$ A1 ft
- Force is 1.18 N A1 **3**

June 2004

GCE A AND AS LEVEL
AICE

MARK SCHEME

MAXIMUM MARK: 50

SYLLABUS/COMPONENT: 9709/06, 0390/06

MATHEMATICS
Paper 6 (Probability and Statistics 1)

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/0390	6

1 (i) $\bar{x}_A = 139$ (138.75) $\sigma_A = 83.1$	B1 B1 2	For the mean For the sd														
(ii) team B smaller standard deviation	B1 B1 dep 2	Independent mark Need the idea of spread SR If team A has a smaller sd then award B1 only for 'teamA, smaller sd'														
2 (i) axes and labels points (3,0) (15,160) (20,320) (35,480) (60,640)	B1 B1 B1 3	For correct uniform scales and labels on both axes, accept Frequency, %CF, Number of people, allow axes reversed, allow halves For 3 correct points All points correct and reasonable graph incl straight lines														
(ii) accept 60 – 70 for straight lines 40 – 70 for curve	M1 A1 2	For subtracting from 640 can be implied For correct answer, reasonably compatible with graph														
3 (i) <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>P(X = x)</td><td>$\frac{11}{36}$</td><td>$\frac{9}{36}$</td><td>$\frac{7}{36}$</td><td>$\frac{5}{36}$</td><td>$\frac{3}{36}$</td><td>$\frac{1}{36}$</td></tr></table>	x	1	2	3	4	5	6	P(X = x)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$	M1 A1 A1 3	For 36 in the uncanceled denominator somewhere, accept decimals eg 0.305 recurring or 0.306 etc For 3 correct probabilities All correct
x	1	2	3	4	5	6										
P(X = x)	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$										
(ii) $E(X) = 1 \times \frac{11}{36} + 2 \times \frac{9}{36} + 3 \times \frac{7}{36} + 4 \times \frac{5}{36} + 5 \times \frac{3}{36} + 6 \times \frac{1}{36} = \frac{91}{36}$	M1 A1 2	For calculation of $\sum xp$ where all probs < 1														
4 (i) $z = \frac{350 - 450}{120}$ = -0.833 % small = $1 - 0.7975 = 0.2025$ or 20.25%	M1 A1 A1 3	For standardising accept 120 or $\sqrt{120}$, no cc For correct z value, + or -, accept 0.83 For answer rounding to 0.202 or 0.203														
(ii) $0.7975 \div 2 = 0.39875$ each $\Phi z_2 = 0.60125$ $z_2 = 0.257$ $x = 120 \times 0.257 + 450$ = 481	M1 M1dep M1 M1dep A1 5	For dividing their remainder by 2 For adding their above two probs together or sub from 1 For finding the z corresponding to their probability For converting to x from a z value For answer, rounding to 481														

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/0390	6

5 (a) (i) $3 \times 5 \times 3 \times 2$ or ${}_3C_1 \times {}_5C_1 \times {}_3C_1 \times 2$ $= 90$	M1 A1 2	For multiplying $3 \times 5 \times 3$ For correct answer
(ii) $(3 \times 5 \times 2) + (3 \times 3) + (5 \times 2 \times 3)$ $= 69$	M1 M1 A1 3	For summing options that show S&M, S&D, M&D $3 \times 5 \times a + 3 \times 3 \times b + 5 \times 3 \times c$ seen for integers a, b, c For correct answer
(b) ${}_{14}C_5 \times {}_9C_5 \times {}_4C_4$ or equivalent $= 252252$	M1 M1 A1 3	For using combinations not all ${}_{14}C_5$... For multiplying choices for two or three groups For correct answer NB $14!/5!5!4!$ scores M2 and A1 if correct answer
6 (i) 	B1 B1 B1 B1 4	For top branches correct (0.65, 0.9, 0.1) For bottom branches correct (0.35, 0.8, 0.2) For win/lose option after 2 nd in (0.6, 0.4) For all labels including final lose at end of bottom branch
(ii) $0.65 \times 0.1 + 0.35 \times 0.8 \times 0.4 + 0.35 \times 2$ $= 0.247$	M1 M1 A1 3	For evaluating 1 st in and lose seen For 1 st out 2 nd in lose, or 1 st out 2 nd out lose For correct answer
(iii) $\frac{0.65 \times 0.1}{0.247}$ $= 0.263 (= 5/19)$	M1 A1ft 2	For dividing their 1 st in and lose by their answer to (ii) For correct answer, ft only on 0.65×0.1 /their (ii)

Page 3	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/0390	6

7 (i) $P(0) = (0.8)^{15} (= 0.03518)$ $P(1) = {}_{15}C_1 \times (0.2) \times (0.8)^{14}$ $(= 0.1319)$ $P(2) = {}_{15}C_2 \times (0.2)^2 \times (0.8)^{13}$ $(= 0.2309)$ $P(X \leq 2) = 0.398$	B1	For correct numerical expression for P(0)
	B1	For correct numerical expression for P(1) or P(2)
	B1 3	For answer rounding to 0.398
 (ii) $1 - (0.8)^n \geq 0.85$ $0.15 \geq (0.8)^n$ $n = 9$	M1	For an equality/inequality involving 0.8, n, 0.85
	M1 dep	For solving attempt (could be trial and error or lg)
	A1 3	For correct answer
 (iii) $\mu = 1600 \times 0.2 = 320$, $\sigma^2 = 1600 \times 0.2 \times 0.8 = 256$ $P(X \geq 290) \text{ or } P(X < 350)$ $= 1 - \Phi\left(\frac{289.5 - 320}{\sqrt{256}}\right) = 1 - \Phi(-1.906)$ $= \Phi(1.906) = 0.972$	B1 M1	For both mean and variance correct For standardising, with or without cc, must have $\sqrt{\quad}$ on denom
	M1	For use of continuity correction 289.5 or 290.5
	M1	For finding an area > 0.5 from their z
	A1 5	For answer rounding to 0.972

June 2004

GCE A AND AS LEVEL

MARK SCHEME

MAXIMUM MARK: 50

SYLLABUS/COMPONENT: 9709/07, 8719/07

**MATHEMATICS AND HIGHER MATHEMATICS
Paper 7 (Probability and Statistics 2)**

Page 1	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	7

1 (i) $H_0: \mu = 15$ or $p = 0.25$ $H_1: \mu > 15$ or $p > 0.25$	B1 1	For H_0 and H_1 correct
(ii) Test statistic $z = \pm \frac{21.5 - 15}{\sqrt{60 \times 0.25 \times 0.75}} = 1.938$ OR test statistic $z = \pm \frac{\frac{22}{60} - \frac{0.5}{60} - \frac{15}{60}}{\sqrt{\frac{0.25 \times 0.75}{60}}} = 1.938$ CV $z = 1.645$ In CR Claim justified	M1 A1 M1 A1ft 4	For attempt at standardising with or without cc, must have $\sqrt{\quad}$ something with 60 in on the denom For 1.94 (1.938) For comparing with 1.645 or 1.96 if 2-tailed, signs consistent, or comparing areas to 5% For correct answer(ft only for correct one-tail test)
2 (i) Mean = $3.5 + 2.9 + 3.1 = 9.5$ Var = $0.3^2 + 0.25^2 + 0.35^2 (=0.275)$ St dev = 0.524	B1 M1 A1 3	9.5 as final answer For summing three squared deviations For correct answer
(ii) $z = \frac{9 - 9.5}{\sqrt{\frac{\text{their var}}{4}}} = -1.907$ or $z = \frac{36 - 38}{\sqrt{(4 \times \text{their var})}} = -1.907$ $\Phi(1.907) = 0.9717 = 0.972$	M1 M1 A1 3	For standardising, no cc For $\sqrt{\frac{\text{their var}}{4}}$ or $\sqrt{(4 \times \text{their var})}$ in denom - no 'mixed' methods. For correct answer
3 (i) $E(2X - 3Y) = 2E(X) - 3E(Y) = 16 - 18 = -2$	M1 A1 2	For multiplying by 2 and 3 resp and subt For correct answer
(ii) Var $(2X - 3Y) = 4\text{Var}(X) + 9\text{Var}(Y)$ $= 19.2 + 54$ $= 73.2$	B1 M1 M1 A1 4	For use of var $(Y) = 6$ For squaring 3 and 2 For adding variances (and nothing else) For correct final answer
4 (i) $\bar{x} = 375.3$ $\sigma^2_{n-1} = 8.29$	B1 M1 A1 3	For correct mean (3.s.f) For legit method involving $n-1$, can be implied For correct answer
(ii) $p = 0.19$ or equiv. $0.19 \pm 2.055 \times \sqrt{\frac{0.19 \times 0.81}{200}}$ $0.133 < p < 0.247$	B1 M1 B1 A1 4	For correct p For correct form $p \pm z \times \sqrt{\frac{pq}{n}}$ either/both sides For $z = 2.054$ or 2.055 For correct answer

Page 2	Mark Scheme	Syllabus	Paper
	A AND AS LEVEL – JUNE 2004	9709/8719	7

5 (i) $\frac{c-54}{3.1/\sqrt{10}} = -1.282$ $c = 54 - 1.282 \times \frac{3.1}{\sqrt{10}} = 52.74$	B1 M1 A1 A1	For + or – 1.282 seen For equality/inequality with their $z (\pm)$ (must have used tables), no $\sqrt{10}$ needed (c can be numerical) For correct expression (c can be numerical, but signs must be consistent) For correct GIVEN answer. No errors seen.
(ii) $P(\bar{x} > 52.74) = 1 - \Phi\left(\frac{52.74 - 51.5}{3.1/\sqrt{10}}\right)$ $= 1 - \Phi(1.265) = 1 - 0.8971$ $= 0.103$ or 0.102	B1 M1 A1 A1	For identifying the outcome for a type II error For standardising, no $\sqrt{10}$ needed For ± 1.265 (accept 1.26-1.27) For correct answer
6 (i) $P(5) = e^{-6} \times \frac{6^5}{5!} = 0.161$	M1 A1	For an attempted Poisson $P(5)$ calculation, any mean For correct answer
(ii) $P(X \geq 2) = 1 - \{P(0) + P(1)\}$ $= 1 - e^{-1.6}(1 + 1.6)$ $= 0.475$	B1 M1 A1	For $\mu = 1.6$, evaluated in a Poisson prob For $1 - P(0) - P(1)$ or $1 - P(0) - P(1) - P(2)$ For correct answer
(iii) $P(1 \text{ then } 4 \mid 5) = \frac{(e^{-3} \times 3) \times (e^{-3} \times \frac{3^4}{4!})}{e^{-6} \times \frac{6^5}{5!}}$ $= 0.156$ or 5/32	M1 M1 A1	For multiplying $P(1)$ by $P(4)$ any (consistent) mean For dividing by $P(5)$ any mean For correct answer
7 (i) $c \int_0^5 t(25 - t^2) dt = 1$ $c \left[\frac{25t^2}{2} - \frac{t^4}{4} \right]_0^5 = 1$ $c \left[\frac{625}{2} - \frac{625}{4} \right] = 1 \Rightarrow c = \frac{4}{625}$	M1 A1 A1	For equating to 1 and a sensible attempt to integrate For correct integration and correct limits For given answer correctly obtained
(ii) $\int_2^4 ct(25 - t^2) dt = \left[\frac{25ct^2}{2} - \frac{ct^4}{4} \right]_2^4 = c[136] - c[46]$ $= \frac{72}{125} \quad (0.576)$	M1* M1*dep A1	For attempting to integrate $f(t)$ between 2 and 4 (or attempt 2 and 4) For subtracting their value when $t = 2$ from their value when $t = 4$ For correct answer
(iii) $\int_0^5 ct^2(25 - t^2) dt = \left[\frac{4}{625} \times \frac{25t^3}{3} - \frac{4}{625} \times \frac{t^5}{5} \right]_0^5$ $= \frac{8}{3}$	M1* A1 M1*dep A1	For attempting to integrate $tf(t)$, no limits needed For correct integrand can have c (or their c) For subtracting their value when $t=0$ from their value when $t=5$ For correct answer